

ScikitLearn 操作記錄單 2

組別: _____ 學號: _____ 姓名: _____

Supervised Learning

1. 請根據以下教學資源操作: <http://www.cse.msu.edu/~ptan/dmbook/tutorials/tutorial6/tutorial6.html>
2. 請自行查詢了解下列 scikit-learn 模組的功能作用 <https://scikit-learn.org/stable/>

Classification	Module	Function	試寫程式，實驗該函式所提供功能及主要參數設定效果
K-Neighbors classification	sklearn.neighbors	KNighborsClassifier ()	
	sklearn.neighbors	KNighborsRegressor ()	
Naïve Bayes Classifiers	sklearn.naive_bayes	Gaussian Naive Bayes()	
	sklearn.naive_bayes	MultinomialNB()	
Decision Trees Classification	sklearn.tree	DecisionTreeClassifier()	
	sklearn.tree	DecisionTreeRegressor()	

SVM Classification	Sklearn.svm	LinearSVC()	
	Sklearn.svm	SVC()	
ANN Classification	Sklearn. neural_network	MLPClassifier()	
	Sklearn. neural_network	MLPRegressor()	
Ensemble classifier	Sklearn.ensemble	RandomForestClassifier ()	
	Sklearn.ensemble	GradientBoostingClassifier() GradientBoostingRegressor()	

Evaluation	Sklearn.model_selection	KFold()	
	Sklearn.model_selection	ShuffleSplit()	
	Sklearn.metrics	confusion_matrix() classification_report() f1_score() precision_recall_curve()	

補充(regression model)

<http://www.cse.msu.edu/~ptan/dmbook/tutorials/tutorial5/tutorial5.html>

其他參考資源:

- machine learning 參考書: "[Introduction to Machine Learning with Python](#)" 之 github code

https://github.com/amueller/introduction_to_ml_with_python/blob/master/02-supervised-learning.ipynb

https://github.com/amueller/introduction_to_ml_with_python/blob/master/05-model-evaluation-and-improvement.ipynb

Scikit Learn documentation(<http://scikit-learn.org/stable/index.html>)

- 尋搜尋其他可信網路資源